BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
HYDERABAD CAMPUS

CS F372: Operating Systems

Headers for Assignment-2 problems

Workload_cpu.h

```
#ifndef WORKLOAD_CPU_H
#define WORKLOAD_CPU_H

#define CPU_ITERATIONS 150000000L

static void cpu_task(int id)
{
   volatile double acc = 0.0;
   long i;

   (void) id;

   for (i = 0; i < CPU_ITERATIONS; i++)
       acc += (double) i * 1.000001;
}

#endif
```

Workload_io.h

```
#ifndef WORKLOAD_IO_H
#define WORKLOAD_IO_H

#include <stdio.h>
#include <string.h>
#include <fcntl.h>
#include <unistd.h>

#define IO_OPERATIONS 2000
#define IO_BLOCK_SIZE 4096

static void io_task(int id)
{
   char path[64];
   char block[IO_BLOCK_SIZE];
   int fd, i;

   snprintf(path, sizeof(path),
"/tmp/io_worker_%d.dat", id);
   memset(block, 'x', sizeof(block));

   fd = open(path, O_CREAT | O_WRONLY | O_TRUNC,
0644);
   if (fd < 0)
       return;

   for (i = 0; i < IO_OPERATIONS; i++) {
       if (write(fd, block, sizeof(block)) !=
sizeof(block))
           break;
```

```
        fsync(fd);
    }

    close(fd);
    unlink(path);
}

#endif
```